Proposal: JUST Regulations Assistant

Team Project, Phase 1: Problem and Data · Building LLM Systems: From Models to Production · Jordan University of Science and Technology (JUST), Irbid

1. Team and roles

Member	Phase 1 role	Responsible for
Hamza Moeen Rasheed Essa	Product lead + Evaluation lead	User and problem section; the 30 test questions, expected answers and scoring
Mohammad Ziad Abdelmajeed Al Rousan	Data lead	Collecting the 36 documents with links, checking extracted Arabic text and OCR numbers, data card
A'sem Zakariya Ali Elfrieh	Engineering lead	GitHub repository, text-extraction pipeline and tests, planned architecture on free compute

Roles rotate at the start of every phase, so each member holds every role by the end of the project.

2. User and problem

User: JUST bachelor students, all majors. **Task:** finding the correct academic rule, for example absence limits, withdrawal deadlines, exam absence and excuses, grades and GPA, probation, postponement, graduation conditions and discipline.

Today: the rules are spread across separate official PDFs. Students open them one by one or ask friends, and a wrong answer can cost them grades, fees or a semester.

Evidence:

- **Fragmentation:** the official regulations page lists **346 separate PDF documents**, by title only, with no search across their contents. "Can I still withdraw?" needs the 17-page degree regulation.
- **Not searchable:** 17 of our 36 documents are scans or use broken fonts, so their text cannot be searched or copied; copying from the other 19 produces **895 corrupted words**.
- **Conflicting rules:** a medical excuse must reach the Health Center within **2 working days** (Medical Excuses Instructions, Art. 4), but the degree regulation allows **2 weeks** to submit it to the dean (Art. 7(e)).

[PLACEHOLDER: real student questions, to be added before submission.] 3 to 5 real questions collected from JUST bachelor students (exact wording, date collected, major; no names), with how many students we asked.

3. What the system will do

A chat assistant for JUST bachelor students. Students ask in Arabic and the assistant answers in Arabic, **only from the official JUST documents**, citing the document and article with every answer. If the answer needs a personal record or a committee decision, or is not in the documents, it says «لا أعرف» ["I don't know"] and names the responsible office.

> **Student:** لحد أي أسبوع بقدر أنسحب من مادة نظرية؟ [Until which week can I withdraw from a theory course?]
> **Answer:** حتى 14 أسبوعاً من بداية الفصل (13 للمساقات العملية، وأول 7 أسابيع في الصيفي). [Up to 14 weeks from the start of the semester (13 for practical courses, the first 7 weeks in summer).]
> Source: Degree Regulation, Art. 12(b).

> **Student:** غبت عن الامتحان الفصلي، شو لازم أعمل؟ [I missed the midterm exam. What should I do?]
> **Answer:** تقدم عذراً لرئيس القسم الذي يطرح المساق خلال أسبوع؛ إذا قُبل يُعقد لك امتحان تكميلي، وإلا توضع لك علامة صفر. [Submit an excuse to the head of the department offering the course within a week; if accepted you get a makeup exam, otherwise a zero.]
> Source: Degree Regulation, Art. 8(b)/9.

> **Student:** هل رح تقبل اللجنة الطبية عذري؟ [Will the medical committee accept my excuse?]
> **Answer:** لا أعرف؛ قبول العذر قرار اللجنة الطبية الطلابية في المركز الصحي، ويمكن الاستعلام عن قرارها من خلال عميد كليتك. [I don't know; this is the Student Medical Committee's decision at the Health Center. You can ask about it through your dean.]

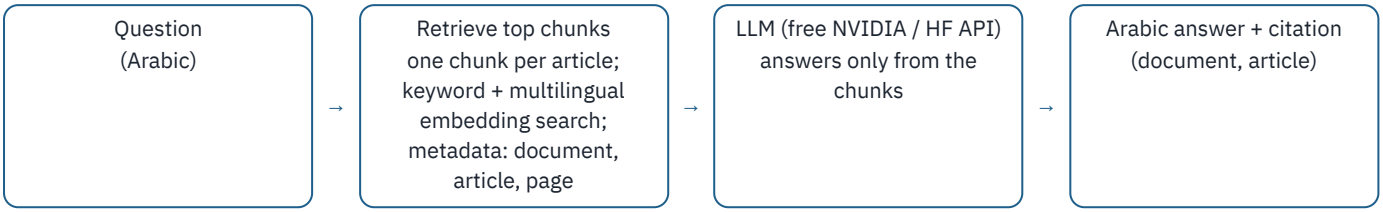
4. Data

Source (official JUST regulations page)	Docs / pages	Status
Student rules: degree regulation, exams, medical excuses, discipline 2024, conduct code, fees, aid and loans, level exams, change of major, insurance, housing, weather suspension, and more	24 / 73	In hand, text extracted

Rules mainly about university staff; a few matter to students (faculty office hours, student evaluation of faculty), the rest are kept but are not the focus	12 / 34	In hand, text extracted
Planned next collection: study plans and course catalogs for all bachelor majors (course codes, credit hours, prerequisites), published on www.just.edu.jo	not yet	Phase 2

In hand: 36 Arabic PDFs, 107 pages, about 80,000 tokens. Details, links and samples: data card and `data/SOURCES.md`.

5. Planned architecture



If retrieval confidence is low, or the question is personal or needs a decision, the system says «لا أعرف» ["I don't know"] and names the office. **Room to grow (RAG + agent):** later phases add tools the agent can call: an absence-limit calculator, a GPA-after-retake calculator, and a study-plan prerequisite checker. These are compared against plain RAG on the same test set.

6. How we will measure success

Correct answer: it states the same rule and every number (weeks, %, hours, GPA) as the expected answer, and nothing that contradicts the documents. **Numbers we report** (computed by `eval/score.py`), overall and by difficulty:

- **Answer accuracy**: manual review, with an LLM judge as a helper that a person checks.
- **Citation accuracy**: the citation names the right document and article.
- **Refusal accuracy**: out-of-scope questions answered «لا أعرف» ["I don't know"], and answerable questions not refused.
- **Retrieval recall@5**: the source article is among the top 5 retrieved chunks.

Baseline: keyword (BM25) retrieval + the same LLM and prompt. Every later version must beat it on the same 30 questions. Seed test set: 30 Arabic questions (15 easy, 10 medium, 5 hard; 3 refusals), answers written from the documents before running any model.

7. Risks

- **OCR errors in numbers and article numbers.** Backup: numbers in the test set come from text-layer pages or were checked against the PDF page image; key tables (grades, fees) are kept as hand-checked CSV.
- **Answers can depend on the major and the study-plan year.** Backup: every document and chunk carries metadata (major, plan year); when an answer depends on them, the system first asks the student's major.
- **Conflicting or outdated rules.** Backup: store the decision date with each chunk; when two rules differ, give both with their sources.
- **A wrong answer can cost grades or fees.** Backup: always cite the source; present the assistant as guidance, not an official decision.
- **Arabic quality and limits of free models; scope.** Backup: compare free models on the test set in Phase 2; add the study plans only after the regulations reach the baseline.